Date	01 October 2026	**Maximum Marks**	4 Marks
Team ID	SWTID-2026-5883	**Document Type**	Solution Requirements
Project Name	**Pocket Smart AI: Your Smart Budget & Recommendation Assistant**		

SOLUTION REQUIREMENTS DOCUMENT

Define Functional and Non-Functional Requirements: Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.
• **Functional Requirements (FR):** Define specific behaviors, functions, or features of the system (What the system should do).
• **Non-Functional Requirements (NFR):** Specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Team Members & Module Allocation

S.No	Student Name	Assigned Module / Requirement Area	Responsibilities & Deliverables
1	**Jashvini.K**	Authentication, Authorization & Security NFRs	Design secure login flows, MFA, user role permissions, access control policies, and end-to-end data privacy framework.
2	**Indhumathi.D**	External Interfaces & Open Banking APIs	Implement integrations with bank APIs, merchant recommendation providers, payment gateways, and data sync layers.
3	**Jayapriya Rajkumar**	Transactions Processing & Performance NFRs	Build core transaction logging, real-time budget tracking, system response speed optimization, and throughput scaling.
4	**Jonal.A**	Reporting, Analytics & Reliability NFRs	Develop interactive spending dashboards, downloadable financial insights, automated budget alerts, and 99.9% availability management.
5	**Kavibharathi.K**	Business Rules, Compliance & Usability NFRs	Define smart AI recommendation budget constraints, ensure regulatory compliance (GDPR/RBI), and enforce WCAG accessibility.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	**Authentication**	Users must securely authenticate using Multi-Factor Authentication (MFA), biometric access (Fingerprint/FaceID), and OAuth 2.0 / JWT token-based login.	**High**
2	**Authorization levels**	Role-Based Access Control (RBAC) defining *Standard User* (personal budget view), *Premium User* (AI automated insights & auto-saving), and *System Admin* (system monitoring and API governance).	**High**
3	**External interfaces**	Seamless API integration with Plaid/Open Banking APIs for automatic expense sync, Merchant Recommendation APIs for personalized deals, and SMS/Push notification gateways.	**High**

S.No	Requirement Category	Requirement Description	Priority
4	**Transactions processing**	Real-time categorization of credit/debit transactions, income vs. expense calculation, automated budget deduction, and anomaly detection for unexpected spending.	**High**
5	**Reporting**	Generate interactive monthly spending reports, category-wise pie charts, historical savings progress bars, and exportable PDF/CSV financial summaries.	**Medium**
6	**Business rules, etc.**	Enforce 50/30/20 budget framework rules; automatically trigger over-budget warning alerts when user spends 80% and 100% of allocated category limits.	**High**
7	**Compliance to laws / regulations**	Strict compliance with Data Privacy laws (GDPR, Digital Personal Data Protection Act) and financial standards (PCI-DSS for transaction handling and RBI data localization).	**High**
8	**Other**	Voice-assisted expense input ("Add $20 for coffee") and offline caching mode to allow local entry logging when internet connection is unavailable.	**Medium**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	System must deliver fast dashboard response times and instant transaction classification.	Dashboard load in < 1.5 seconds; API latency < 200ms.
2	**Scalability**	Architecture must scale dynamically during peak financial query periods or salary credit days.	Supports up to 50,000 concurrent active users with horizontal auto-scaling.
3	**Security & Data Privacy**	All sensitive financial data and user credentials must be protected at rest and in transit.	AES-256 encryption at rest, TLS 1.3 in transit, and quarterly penetration testing.
4	**Reliability & Availability**	The solution must ensure continuous service availability with failover backup systems.	99.9% uptime per month (maximum allowed downtime < 43 mins/month).
5	**Usability & Accessibility**	Clean mobile-first UI with dark/light mode and accessible options for visually impaired users.	Complies with WCAG 2.1 AA standards; System Usability Scale (SUS) score > 85.
6	**Maintainability & Portability**	Modular microservices architecture allowing cross-platform mobile and web application support.	100% cloud-native (Docker/ Kubernetes); operational across iOS, Android, and Web browsers.